Lecture 13

February 19, 2026

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Why Futures Markets Exist

- Futures contracts are standardized agreements to buy or sell an asset at a future date for a predetermined price.
- They serve three primary economic functions:
  - Risk Transfer
    - Producers hedge price risk (e.g., oil, wheat, metals)
    - Consumers hedge input cost risk
    - Financial institutions hedge interest rate, currency, or equity exposure
  - Price Discovery
    - Futures prices incorporate market expectations
    - High liquidity concentrates information
    - Forward curve reflects cost-of-carry and market risk premia
  - Capital Efficiency
    - No full upfront payment required
    - Margin-based collateralization
    - Embedded leverage
- In equilibrium:
  - Hedgers transfer risk
  - Speculators absorb risk
  - Arbitrageurs enforce pricing relationships

Forward Contracts vs Futures Contracts

- Forward Contract
  - Private OTC agreement
  - Custom terms (size, maturity, underlying)
  - Settlement at maturity
  - Counterparty credit risk
- Futures Contract
  - Exchange-traded and standardized
  - Daily marking to market
  - Clearinghouse guarantees performance
  - Margin requirements
- Key Structural Difference:
  - Forwards settle once
  - Futures settle daily
  - Daily settlement reduces credit exposure but introduces reinvestment effects.

Basic Futures Contract Structure

- A futures contract is a standardized agreement specifying all economically relevant terms in advance.
- Each contract defines:
  - Underlying Asset
    - Agricultural commodities (wheat, corn, soybeans, cattle)
    - Metals & energy (gold, copper, crude oil, natural gas)
    - Interest rate instruments (Treasury bonds, Eurodollars, SOFR)
    - Currencies (EUR/USD, JPY/USD, GBP/USD)
    - Equity indexes (S&P 500, NASDAQ-100, Russell 2000)
    - Etc.
    - Commodities can also have specific quality / grade specifications

Basic Futures Contract Structure (con.)

- Contract Size
  - Fixed quantity of the underlying
  - Example:
    - Crude oil: 1,000 barrels
    - Treasury bond futures: $100,000 face value
    - S&P 500 futures: $250 × index level (standard contract)
- Delivery Date(s)
  - Specific calendar months (e.g., March, June, September, December)
  - Often called “delivery cycle”
- Settlement Procedure
  - Physical delivery (many commodities)
  - Cash settlement (equity indexes, some rates)

Basic Futures Contract Structure (con.)

- Tick Size (Minimum Price Movement)
  - Crude oil: $0.01 per barrel → $10 per contract
  - Treasury bond futures: 1/32 of par → $31.25 per tick
  - S&P 500 futures: 0.25 index points → $62.50 per tick
- Standardization increases:
  - Liquidity
  - Transparency
  - Price comparability
  - Arbitrage efficiency
- The exchange defines every term so that contracts are interchangeable and actively tradable.

Margining System

- Futures contracts require collateral, not full payment.
- Because futures are zero-cost at initiation, the exchange uses a margin system to ensure performance.
- Key components:
  - Initial Margin
    - Deposit required when the position is opened
    - Acts as a performance bond, not a down payment
    - Typically a small fraction of notional value (5–15%)
  - Maintenance Margin
    - Minimum margin account balance required to keep the position open
    - Usually below initial margin
    - If balance falls below this level → margin call
  - Variation Margin
    - Daily cash flows from marking to market
    - Gains credited, losses debited each day
- Important distinction:
  - Margin is not borrowed money
  - Futures create embedded leverage because:
    - Notional Exposure >Margin Posted

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Marking to Market

- A defining feature of futures is daily settlement.
- At the end of each trading day:
  - Daily Gain/Loss = $(F_t - F_{t-1}) \times Q$
  - Where:
    - F_t = today's settlement futures price
    - F_{t-1} = yesterday's settlement price
    - Q = contract size
- Mechanics:
  - If futures price rises → long receives cash, short pays
  - If futures price falls → short receives cash, long pays
- Thus:
  - Futures positions are "reset" daily
  - Contract value is driven back toward zero each day
  - Losses must be funded immediately
  - Marking to market converts a long-dated forward-style payoff into a sequence of short-horizon cash flows.

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The Role of the Clearinghouse

- Futures exchanges eliminate bilateral counterparty risk through a clearinghouse.
- The clearinghouse:
  - Becomes the buyer to every seller
  - Becomes the seller to every buyer
  - Guarantees contract performance
- This process is called novation.
- Instead of exposure to another trader, each participant faces the clearinghouse.
- Risk management tools:
  - Initial and variation margin
  - Daily settlement
  - Position limits
  - Default funds
- The clearinghouse is the institutional mechanism that allows futures markets to scale safely.

Economic Consequences of Clearing

- Clearinghouse structure produces major economic effects:
  - Reduces Counterparty Credit Risk
    - Losses are collateralized daily
    - Default exposure is short-lived
  - Enhances Liquidity
    - Traders do not need to evaluate counterparty quality
    - Standardized contracts trade continuously•
  - Improves Price Efficiency
    - Arbitrage can be executed with minimal friction
    - Futures prices closely track no-arbitrage bounds
  - However:
    - Risk becomes concentrated in a central institution
    - Clearinghouses become systemically important
    - Stress events require careful margin and liquidity management
- Thus, futures markets are among the most regulated (CFTC, NFA) and infrastructure-critical financial markets.

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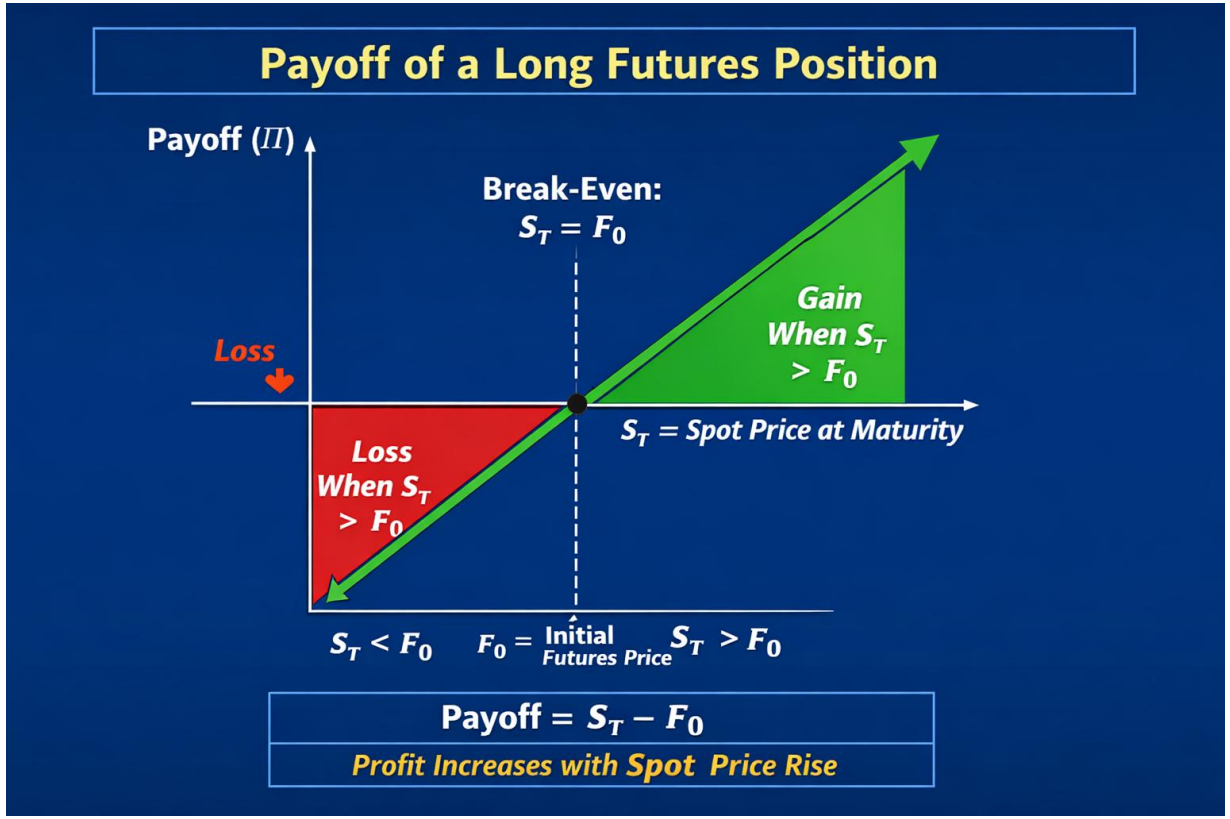
Payoff of a Long Futures Position

- A long futures position commits to buy the underlying at maturity for F_0.
- At maturity T:
  - $\text{Payoff}_{\text{Long}} = S_T - F_0$
- Where:
  - S_T = spot price at maturity
  - F_0 = futures price when contract was initiated
- Properties:
  - Linear exposure to the underlying
  - Gains when prices rise
  - Losses when prices fall
  - Symmetric payoff structure
  - No upfront purchase cost



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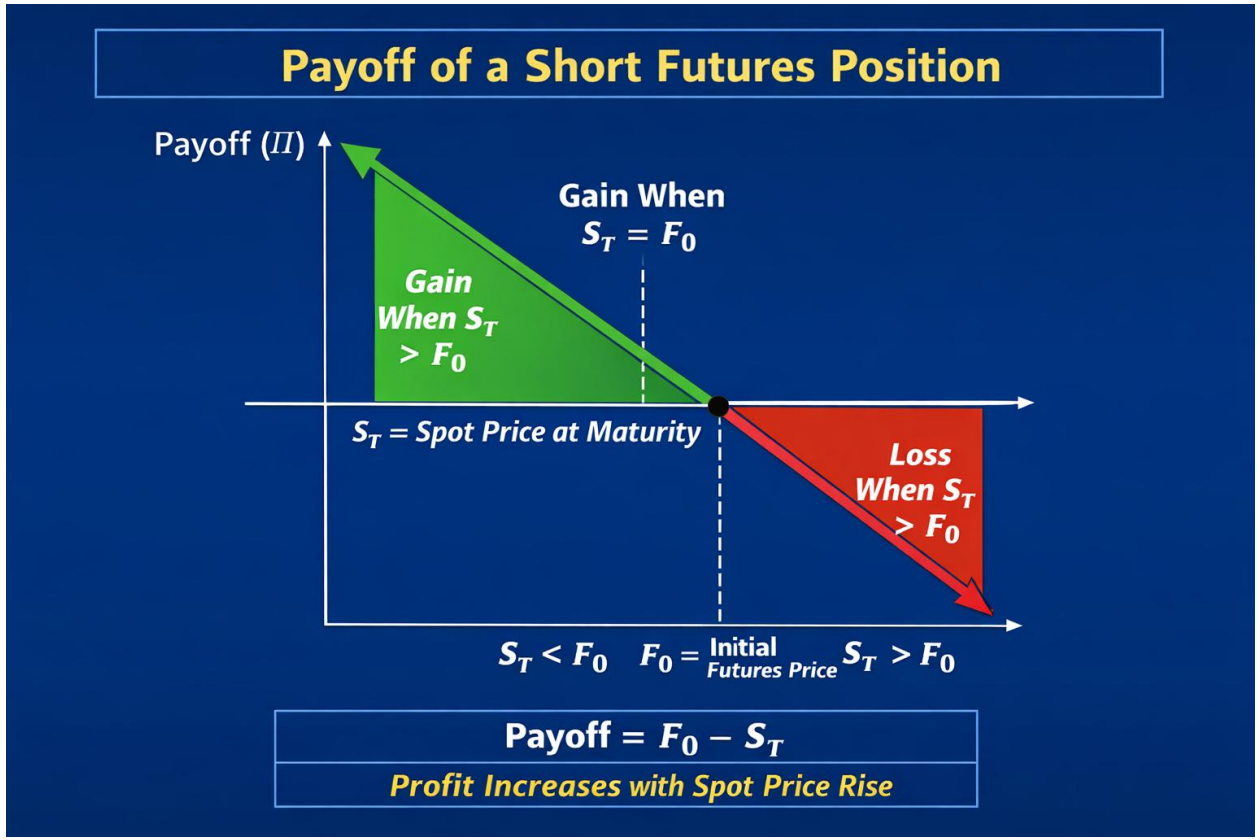
Payoff of a Short Futures Position

- A short futures position commits to sell the underlying at maturity for F_0.
- At maturity T:
  - $\text{Payoff}_{\text{Short}} = F_0 - S_T$
- Properties:
  - Linear payoff with slope −1
  - Gains when prices fall
  - Losses when prices rise
  - Mirror image of the long payoff
  - No initial cash inflow beyond margin



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Futures Pricing Question

- A futures contract entered at time 0 specifies a delivery price: F_0=futures price agreed upon today for exchange at time T
- Key institutional feature:
  - Entering a futures contract requires no upfront purchase payment
  - Only margin collateral is posted to guarantee performance
  - The contract is initiated at zero net cost
  - At maturity T, the payoff to a long futures position is:

$$\Pi_T = S_T - F_0$$

- Because no asset is purchased at time 0, what determines the equilibrium futures price F_0?
  - The futures price cannot be arbitrary:
    - Spot and futures markets are jointly traded
    - Investors can borrow, lend, and trade the underlying asset
    - These opportunities impose a strict pricing relationship between S_0 and F_0

Replication: Buying the Asset vs. Using Futures

- To determine the futures price F_0, compare two ways to obtain exposure to owning the asset at time T.
  - Strategy Part A: Buy the Asset Today (Cash Market)
    - At time 0:
      1. Borrow S_0 at the risk-free rate r
      2. Purchase the underlying asset in the spot market
    - Cash flow at initiation: $+S_0$ (borrow)$-S_0$ (buy asset) $= 0$
    - At maturity T:
      - Asset is worth S_T
      - Loan repayment equals: $S_0 e^{rT}$
    - Net payoff: $S_T - S_0 e^{rT}$

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Replication: Buying the Asset vs. Using Futures (con.)

- Strategy Part B: Long Futures Contract
  - Enter a long futures contract with delivery price F_0.
  - At maturity T, payoff is: $S_T - F_0$
- No-Arbitrage Pricing Condition
  - Both strategies produce the same exposure to S_T.
  - Therefore, their payoffs must be equal: $S_T - F_0 = S_T - S_0 e^{rT}$
  - Cancel S_T: $F_0 = S_0 e^{rT}$
- This is the fundamental spot–futures parity relationship for a non-income asset.

If Futures Are Overpriced: Cash-and-Carry Arbitrage

- Suppose the futures price exceeds the cost of replicating the payoff:

$$F_0 > S_0 e^{rT}$$

- Arbitrage Strategy
  - At time 0:
    - Borrow S_0 at rate r
    - Buy the asset in the spot market
    - Enter a short futures position
    - Net initial cash flow: 0
  - At maturity T:
    - Deliver asset into futures contract
    - Receive F_0
    - Repay loan $S_0 e^{rT}$
  - Riskless Profit $= F_0 - S_0 e^{rT} > 0$
  - Arbitrageurs sell futures and buy spot, thus futures price falls.

If Futures Are Underpriced: Reverse Arbitrage

- Now suppose:

$$F_0 < S_0 e^{rT}$$

- Reverse Strategy
  - At time 0:
    - Short-sell the asset
    - Invest proceeds S_0 at rate r
    - Enter a long futures position
    - Net initial cash flow: 0
  - At maturity T:
    - Investment grows to $S_0 e^{rT}$
    - Purchase asset via futures at price F_0
    - Return asset to close short position
  - Riskless Profit = $S_0 e^{rT} - F_0 > 0$
  - Arbitrageurs buy futures and short spot, thus futures price rises.

Spot–Futures Parity: Economic Interpretation

- From replication: $F_0=S_0e^{rT}$
- This relationship reflects:
  - The ability to buy the asset today
  - The ability to finance that purchase at the risk-free rate r
  - The fact that a futures contract defers payment until maturity
- What determines the futures price?
  - The futures price is an arbitrage price, pinned down by:
    - Spot price S_0
    - Financing conditions over [0,T]
    - Trading opportunities in spot and futures markets
  - It is not chosen by convention or expectation.
  - Futures Prices Are Not Forecasts
    - In general: $F_0 \neq E[S_T]$
    - The futures price is determined by no-arbitrage, not by beliefs about the future spot price.
    - Risk premia may cause expected spot prices to differ from futures prices.

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Convergence at Maturity

- We also see convergence at maturity
- At the contract's maturity date T, the futures contract requires:
  - Physical delivery of the underlying asset, or
  - Cash settlement based on the prevailing spot price
- Therefore, at expiration:

$$\lim_{t \to T} F_t = S_T$$

- This also makes sense as arbitrageurs could sell one and buy the other if they did not

Futures on Assets with Known Income

- Suppose the underlying asset pays known cash income during [0,T].
- Examples:
  - Fixed coupon payments (bond futures)
  - Known dividends with announced dates
  - Lease payments
- If income of present value PV(Income) is received before maturity, then the cost of carrying the asset is reduced.
- Replication Adjustment
  - When buying the asset today:
    - You pay S_0
    - You receive income during the life of the contract
    - Thus the effective cost of acquiring and carrying the asset is: $S_0 - PV(Income)$
  - No-Arbitrage Futures Price: $F_0 = (S_0 - PV(Income))e^{rT}$
- Income lowers the futures price because the holder of the spot asset receives the cash flows.

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Continuous Dividend Yield (Equity Index Futures)

- For equity indexes, dividends are uncertain but can be modeled as a continuous yield q.
  - Holding the index yields dividends at rate q
  - This reduces the effective cost of carrying the asset
- Futures Pricing Formula

$$F_0=S_0e^{(r-q)T}$$

  - Where:
    - r = risk-free rate
    - q = continuous dividend yield
- If dividend yield exceeds financing cost: $q>r$
- Then: $F_0<S_0$
- Equity index futures often trade below spot when dividend yields are high relative to interest rates.

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Commodities: Storage Costs and Convenience Yield

- For commodities, additional carry components arise.
- Carrying physical inventory involves:
  - Storage costs (u)
  - Insurance and handling
  - Financing costs
- But holding physical inventory may provide benefits:
  - Ability to meet unexpected demand
  - Avoid production disruptions
  - Strategic inventory value
  - This benefit is called the convenience yield (y).
- Generalized Commodities Futures Pricing Formula
  - $F_0=S_0e^{(r+u-y)T}$
  - Where:
    - u = storage cost rate
    - y = convenience yield

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The Cost-of-Carry Model

- Combining all components:

$$F_0=S_0e_{(r+u-y-q)T}$$

- Depending on the asset class:
  - Financial assets: typically u=0
  - Commodities: q=0
  - Equity indexes: u=y=0
- Cost-of-Carry Components:
  - Financing cost r
  - Storage cost u
  - Income yield q
  - Convenience yield y
- The futures price reflects the net cost of carrying the asset until maturity.

Contango and Backwardation (Carry-Based Definition)

- One common usage of contango/backwardation refers to the shape of the futures curve relative to spot today.
  - From cost-of-carry: $F_0=S_0e^{(r+u-y-q)T}$
  - Thus the sign of F_0-S_0 depends on net carry: $r+u-y-q$

Contango (spot-relative)

- $F_0>S_0$
- Occurs when:
  - $r+u > y+q$
- Typical in:
  - Financial futures
  - Commodities with high storage cost
  - Abundant inventory environments

Backwardation (spot-relative)

- $F_0<S_0$
- Occurs when:
  - $y+q>r+u$
- Typical in:
  - Scarce commodities
  - High convenience yield environments
  - High dividend yield equities

Spot, Futures, and Expectations: Q vs P

- Arbitrage determines the futures price: $F_0=S_0e^{(r+u-y-q)T}$
  - This relationship:
    - Requires no beliefs about future spot prices
    - Holds regardless of expectations
    - Is enforced by cash-and-carry arbitrage
- Risk-Neutral Measure (Q)
  - Under the risk-neutral measure: $F_0=E^Q[S_T]$
  - Thus $E^Q[S_T]= S_0e^{(r+u-y-q)T}$
- Physical Measure (P)
  - Under the real-world measure: $E^P[S_T] \neq F_0$
  - The difference reflects a risk premium: $E^P[S_T] - F_0$ =Risk Premium

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Expected Futures Returns and Beta

- A long futures position has payoff:

$$\Pi_T = S_T - F_0$$

- Expected payoff:

$$E^P[\Pi_T] = E^P[S_T] - F_0$$

- Asset pricing implies:
- $E^P[\Pi_T] \propto \beta_{futures}$
  - $\beta > 0$: long futures requires compensation
  - $\beta < 0$: long futures provides hedging value
- Thus futures earn premia depending on their systematic risk.

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Normal Backwardation (Physical-Expectation Sense)

- Keynes/Hicks hedging pressure hypothesis:
  - Natural hedgers are often producers
  - They hedge by shorting futures
  - Speculators take the long side
- Equilibrium implies:

$$F_0 < E^P[S_T]$$

  - So:

$$E^P[S_T - F_0] > 0$$

- Long futures earn positive expected returns.
- This is called normal backwardation.

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Consumer Hedging Pressure (Physical "Contango")

- In some markets, natural hedgers are consumers:
  - Airlines hedging jet fuel
  - Manufacturers hedging metals
- They hedge by going long futures.
- Equilibrium implies:

$$F_0 > E^P[S_T]$$

  - So:

$$E^P[S_T - F_0] < 0$$

- Long futures provide hedging value and may have negative expected returns.

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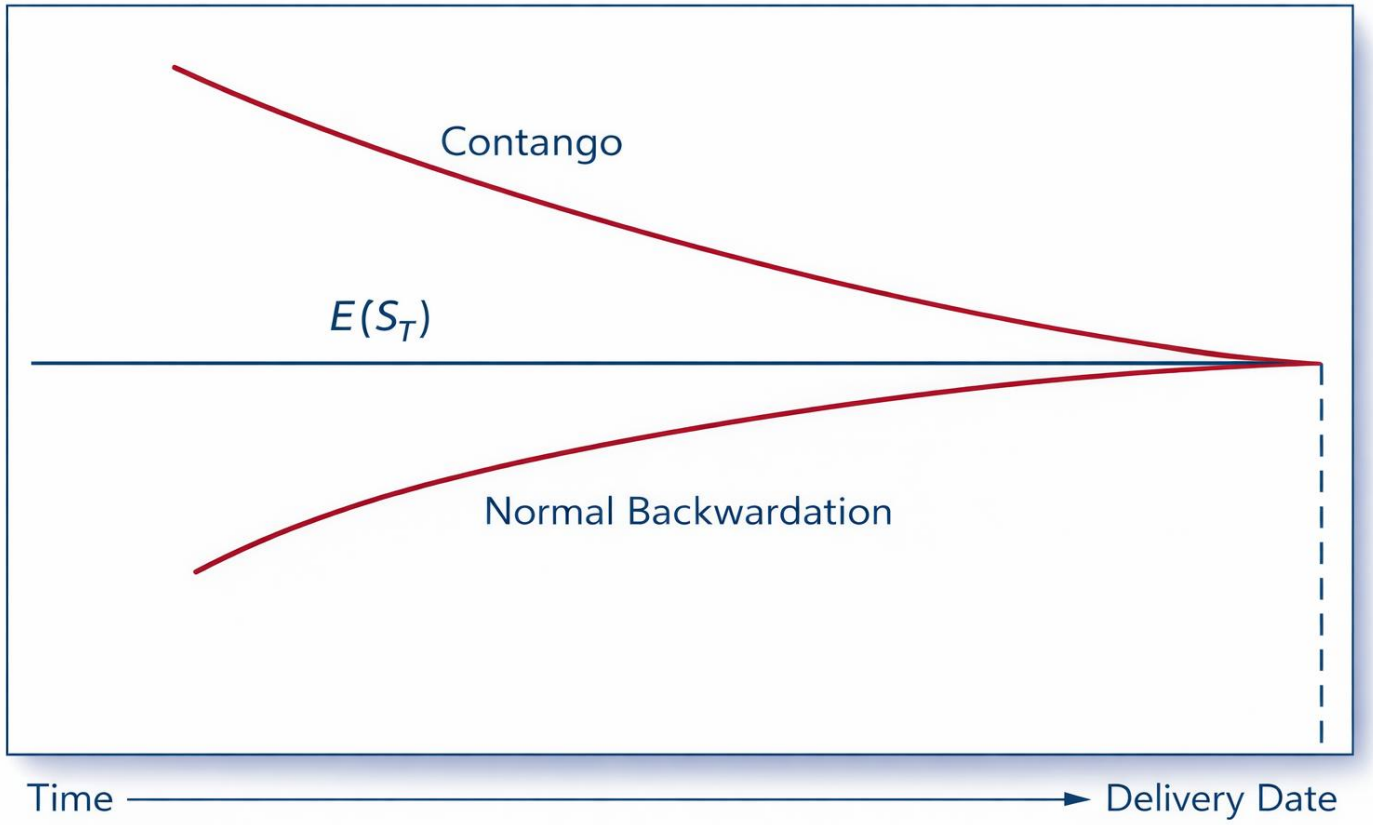
Risk-Premium Diagram: Futures vs Expected Spot

- This figure is expectation-relative:
  - Normal backwardation:

    $F_0 < E^P[S_T]$
  - Consumer-pressure contango:

    $F_0 > E^P[S_T]$
  - It is not a carry diagram.



Two Distinct Uses of the Same Terms

- Contango/backwardation are used with two reference points:
- Spot-Relative (Curve Shape)

$$F_0 \gtreqless S_0$$

  - Driven by net carry: r+u−y−q
- Expectation-Relative (Risk Premium)

$$F^0 \gtreqless E^P[S^T]$$

  - Driven by:
    - Hedging pressure
    - Systematic beta
    - Risk premia
- Arbitrage determines futures prices.
- Asset pricing determines expected futures returns.

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Forward vs Futures: Pricing Equivalence

- Forwards and futures have identical terminal payoff:

$$\Pi_T = S_T - F_0$$

- Under deterministic interest rates:

$$F_0 \text{forward} = F_0 \text{futures}$$

- Reason:
  - Daily settlement does not change present value
  - Reinvestment occurs at known rate
  - No covariance between interest rates and underlying
- Thus, pricing formulas derived earlier apply equally to forwards and futures <u>when rates are non-random</u>.

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Marking-to-Market: Timing of Cash Flows

- A key difference between futures and forwards is the timing of settlement.
  - Forward contracts settle only at maturity T
  - Futures contracts are marked to market daily:

$$\Delta \text{Margin}_t = (F_t - F_{t-1})Q \text{ ; where } Q = \text{quantity}$$

- Thus a futures position generates intermediate cash flows prior to expiration.
- Asset Pricing Implication
  - Because cash flows occur before maturity, their present value depends on:
    - The path of interest rates
    - The covariance between rates and the underlying price
    - Therefore, futures and forward prices coincide only under special conditions.

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Pricing Equivalence Under Deterministic Rates

- If the short rate is non-random over [0,T]:

$$F_0^{\text{futures}} = F_0^{\text{forward}}$$

- Reason:
  - Intermediate settlement cash flows can be reinvested at a known rate
  - Marking-to-market has no valuation effect
- Thus, with deterministic rates, futures and forwards share the same arbitrage price:

$$F_0 = S_0 e^{(r+u-y-q)T}$$

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Stochastic Rates Break Equivalence

- When interest rates are stochastic, daily settlement matters.
- Futures gains/losses are realized earlier and reinvested at uncertain rates.
- Thus: $F_0^{\text{futures}} \neq F_0^{\text{forward}}$
- The difference arises from the covariance between:
  - the underlying price S_t
  - the short rate r_t
- Direction of Bias
  - If: Cov(St,rt)>0 then: $F_0^{\text{futures}} > F_0^{\text{forward}}$
  - If: Cov(St ,rt)<0 then: $F_0^{\text{futures}} < F_0^{\text{forward}}$
  - This is the convexity adjustment.

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Convexity Bias: Economic Intuition

- Consider a long futures position.
- If asset prices tend to rise when rates rise:
  - Gains occur when reinvestment rates are high
  - Futures become more valuable relative to forwards
  - Thus: $F_0^{\text{futures}} > F_0^{\text{forward}}$
- If asset prices fall when rates rise:
  - Losses occur when funding costs are high
  - Futures become less valuable
  - Thus: $F_0^{\text{futures}} < F_0^{\text{forward}}$
- Convexity bias is not an arbitrage violation.
- It reflects:
  - stochastic discounting
  - cash-flow timing
  - covariance structure
- It is most relevant for interest rate futures.

Hedging with Futures: Objective

- Futures contracts are primarily used to reduce risk exposure.
- Let:
  - S_t = spot price of the asset being hedged
  - F_t = futures price
  - h = number of futures contracts per unit of spot exposure
- A hedged position has value change: $\Delta V = \Delta S - h\Delta F$
- Hedging Goal
  - Choose h to reduce variance:

$$\min_h \; Var(\Delta V)$$

- This is a mean–variance hedging problem.

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Minimum-Variance Hedge Ratio

- The optimal hedge ratio solves: $\min_h Var(\Delta S - h\Delta F)$
- The solution is: $h^* = \frac{Cov(\Delta S, \Delta F)}{Var(\Delta F)}$
- Equivalently: $h^* = \rho \frac{\sigma_S}{\sigma_F}$
- Where:
  - ρ = correlation between spot and futures changes
  - σ_S, σ_F = volatilities
- Interpretation: $h*$ is the regression coefficient in: $\Delta S = \alpha + h\Delta F + \varepsilon$

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Optimal Number of Futures Contracts

- The minimum-variance hedge ratio h^* determines the optimal hedge in exposure units.
- In practice, hedgers must choose an integer number of futures contracts.
  - Let:
    - Q_A = size of the spot exposure being hedged(e.g., barrels, bushels, portfolio value)
    - Q_F = contract size of one futures contract(units per contract)
    - h^* = optimal hedge ratio
  - Optimal Number of Contracts: $$N^* = h^* \frac{Q_A}{Q_F}$$
    - Where:
      - N^* = number of futures contracts
      - Long or short depending on hedge direction

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Optimal Number of Futures Contracts (con.)

- Short Hedge (Producer)
  - Exposure: long spot asset → hedge by short futures
  - $N^* > 0 \Rightarrow$ Short N^* contracts
- Long Hedge (Consumer)
  - Exposure: future purchase requirement → hedge by long futures
  - $N^* > 0 \Rightarrow$ Long N^* contracts
- The hedge ratio determines the fraction of the exposure to hedge.
- The contract conversion scales it into tradable contract units.
- This formula applies generally across:
  - commodities
  - currencies
  - equity indexes
  - interest rate futures

Perfect Hedging and Basis Risk

- If spot and futures prices move one-for-one:

$$\Delta S = \Delta F$$

- Then:
  - $h^* = 1$
  - and the hedge eliminates risk.
- In practice, correlation is imperfect.
- Define the basis:
  - $b_t = S_t - F_t$
- Then:
  - $\Delta V = \Delta b_t$
- Residual hedging risk is basis risk.

Basis Convergence and Hedge Horizon

- At expiration:

$$F_T = S_T$$

- Thus:

$$b_T = 0$$

- Basis risk disappears only if:
  - Hedge horizon matches futures maturity
- If the hedge is closed early:

$$b_t \neq 0$$

- Basis fluctuations generate residual risk.
- This motivates contract selection and rolling strategies.

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Cross Hedging

- Often the exact underlying is not traded in futures markets.
  - Example:
    - Jet fuel exposure hedged with crude oil futures
    - Corporate bond exposure hedged with Treasury futures
  - This is cross hedging.
- Optimal hedge ratio becomes:

$$h^* = \rho \frac{\sigma_{\text{Exposure}}}{\sigma_{\text{Futures}}}$$

- Cross hedging increases basis risk because $\rho<1$.

Numerical Example: Minimum-Variance Hedge Ratio

- Goal: hedge an exposure using futures by minimizing variance:

$$\Delta V = \Delta S - h\Delta F$$

- Minimum-variance hedge ratio: $h^* = \rho \frac{\sigma_S}{\sigma_F}$

- Example (Weekly Price Changes)
  - Spot exposure: Jet fuel price index S
  - Hedge instrument: Crude oil futures F
  - Assume historical estimates:
    - Spot volatility: $\sigma_S = 3.0\%$
    - Futures volatility: $\sigma_F = 2.5\%$
    - Correlation: $\rho = 0.80$

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Numerical Example: Minimum-Variance Hedge Ratio (con.)

- Compute Optimal Hedge Ratio:

$$h^* = 0.80 \cdot \frac{0.030}{0.025} = 0.80 \cdot 1.2 = 0.96$$

- Interpretation:
  - A minimum-variance hedge offsets approximately 96% of the spot exposure using futures.
  - Imperfect correlation implies some basis risk remains

Numerical Example: Optimal Number of Contracts

- To implement the hedge, convert h^* into contracts.
- Optimal number of futures contracts: $N^* = h^* \frac{Q_A}{Q_F}$
- Where:
  - Q_A = size of exposure
  - Q_F = contract size
- Exposure and Contract Size
  - Suppose the firm needs to hedge: QA =1,000,000 gallons of fuel-equivalent exposure.
  - One futures contract represents: QF=42,000 gallons-equivalent.

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Numerical Example: Optimal Number of Contracts (con.)

- Compute Number of Contracts:

$$N^* = 0.96 \cdot \frac{1{,}000{,}000}{42{,}000} = 0.96 \cdot 23.81 = 22.86$$

- Thus, the hedge requires approximately 23 contracts
- Direction of Hedge
  - Since the firm is exposed to rising fuel costs (a future buyer), take a long hedge (buy futures)

Hedging Equity Portfolios with Index Futures

- Equity index futures provide rapid adjustment of systematic exposure.
- Let:
  - V_P = portfolio value
  - β_P = portfolio beta
  - F_0 = futures price
  - Contract multiplier = M
- To hedge market risk: $$N^* = \frac{\beta_P V_P}{F_0 M}$$
- Index futures allow:
  - Beta reduction without selling securities
  - Efficient portfolio rebalancing
  - Tactical asset allocation
- Futures are "beta management instruments."

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Duration Hedging with Bond Futures

- Bond futures can hedge interest rate risk.
- Let:
  - D_P = portfolio duration
  - V_P = portfolio value
  - D_F = futures contract duration
  - F_0M = futures notional value
- Duration hedge ratio: $$N^* = \frac{D_P V_P}{D_F(F_0 M)}$$
- Treasury futures provide:
  - Efficient duration adjustment
  - Hedging of yield curve exposure
  - Foundation for swap hedging

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Tailing the Hedge (Present Value Adjustment)

- A futures hedge generates gains/losses through daily settlement.
- Thus the hedge offsets the future value of the exposure, not its present value.
- For exposures occurring at time T, the hedge should be "tailed":
  - If the spot exposure is realized at maturity: Q_A at time T
  - Then its present value is: $Q_A e^{-rT}$
  - A full notional hedge may over-hedge because futures gains are reinvested.

- The tailed number of contracts is: $N^*_{\text{tailed}} = N^* e^{-rT}$

- Equivalently: $N^*_{\text{tailed}} = h^* \frac{Q_A}{Q_F} e^{-rT}$

- Discount the exposure because futures are marked-to-market
- Most relevant for interest rate futures and longer horizons
- Often used in duration hedging and fixed income risk management

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